

Government Savings

Jira / qTest / CI Harmonization — Technical Plan

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Jira Qtest Ci Harmonization Plan

Jira / qTest / CI Harmonization Plan

****As of:**** 2026-07-21

****Scope:**** Government Savings coverage intelligence (read-only MVP)

****Principle:**** Smallest practical solution — extend existing Python utilities; no new application

1. Problem statement

Leadership needs trustworthy, platform-specific coverage intelligence. Today Jira, qTest, Git repositories, GitLab, GitHub Actions, and Jenkins each hold partial truth. They are ****not harmonized**** because:

- | Gap | Evidence | Impact |
- |-----|-----|-----|
- | No common identifier across systems | Sparse qTest refs in test code; Jira keys inconsistent | Cannot auto-reconcile |
- | Stale qTest data | Last export 2026-06-29 (744 cases) | V2 metrics marked Stale |
- | Missing row-level qTest mappings | UP reconciliation ledger | Unmatched records in prototype |
- | Incomplete Jira AC linkage | Jira MCP blocked; export partial | Denominator disputes |
- | Shared repos, different counting units | TestNG methods vs qTest cases vs endpoints | Cannot blend percentages |
- | GitLab / GHA / Jenkins independent | Separate schedulers and artifacts | No single execution view |
- | No normalized central register | Multiple CSVs in KB | Manual leadership synthesis |
- | No freshness SLA | Ad hoc exports | Stale metrics presented as current |
- | No contradiction detection | Manual reviews only | Silent drift |
- | No single governance owner | RACI spread | Threshold and exception delays |

2. Target operating model

Nine lightweight layers — ****no database required for MVP****:

- | # | System | Role |

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|---|-----|-----|

| 1 | **Jira** | Approved scope: epics, stories, acceptance criteria, capability, service/endpoint IDs |
| 2 | **qTest** | Test inventory, manual/automated status, regression designation, requirement
links, execution history |
| 3 | **Git repositories** | Automated implementation, stable test IDs, suite placement, endpoint
mappings |
| 4 | **GitLab / GHA / Jenkins** | Live execution evidence, schedules, pass/fail, merge/deploy
behavior, artifacts |
| 5 | **JaCoCo / SonarQube** | Source-code coverage (metric A), changed-code, thresholds |
| 6 | **Python utilities** | Read-only collectors, normalization, reconciliation, contradiction
detection, reports |
| 7 | **Central versioned register** | CSV/JSON with numerator, denominator, timestamp, source,
confidence, owner, freshness, status |
| 8 | **Reporting** | Leadership summary, technical workbook, exception report, trend report |
| 9 | **Governance** | Approved denominator, owner, cadence, SLA, exception process, bypass
group, audit trail |

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****Metric separation (mandatory):****

- ****A**** Application source-code coverage (JaCoCo/Sonar)
- ****B**** Business automation coverage (scenarios, endpoints, qTest)
- ****C**** Execution coverage (recently ran)
- ****D**** CI integration coverage (in pipeline)
- ****E**** Gate coverage (blocks merge/deploy/release)

3. Current integration inventory

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| Integration | Status | Objects available | Blocker |

|-----|-----|-----|-----|

| Jira MCP / REST | Blocked | Stories, epics, links (when credentialed) | MCP timeout; no token |
| qTest REST | Not configured | Cases, runs, requirements | `QTEST_*` env unset |
| qTest exports | Partial | 744-case PDF snapshot | Stale 2026-06-29 |

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GitLab API / glab	Blocked	Pipelines, schedules, MR status	HTTP 401 expired token
GitHub API / gh	Not configured	Actions workflows, runs	`gh` not installed
Jenkins API	Partial	KB console logs, job names	No live API URL/token
Repository scanners	Available	Test files, suite XML, endpoint tests	Local clone only
Python utilities	Available	UP rebuild, GS deliverables, live validation	Extend for collectors
JaCoCo reports	Partial	Service POMs; no central ingest	Per-repo only
SonarQube	Disabled	N/A on reviewed CI	`RUN\_SONARQUBE: false`

4. Harmonization workflow (MVP)

Weekly automated cycle (read-only)

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1. Jira collector → jira-scope-snapshot.json (scoped stories + AC count)
2. qTest collector → qtest-snapshot.json (cases, automation flag, last run)
3. Repo scanner → automation-test-inventory.csv (implementation)
4. GitLab collector → execution-snapshot.json (last pipeline per job)
5. GHA collector → gha-execution-snapshot.json (when credentialed)
6. Jenkins collector → jenkins-execution-snapshot.json (when credentialed)
7. JaCoCo ingest → source-coverage-snapshot.csv (pilot services)
8. Reconciliation engine → cross-system-traceability.csv + contradictions
9. Register merge → verified-metrics-register.csv (human-approved rows only)
10. Report generator → leadership-summary.md + exception-report.csv

...

Identifier standard (pilot)

| Field | Format | Example |

|-----|-----|-----|

| `automation\_id` | `{platform}-{module}-{scenario}` | `M2-BANKS-GET-LIST` |

| `qtest\_case\_id` | qTest numeric ID | `1234567` |

| `jira\_key` | Standard Jira key | `QA-1405` |

| `endpoint\_key` | HTTP method + path | `GET /mobile2api/v1/banks` |

Embed in test class `@Description`, TestNG `description`, or suite XML comment — **no qTest writes**.

5. Why systems are not one harmonized system today

1. **Different purposes:** Jira tracks delivery scope; qTest tracks test inventory; repos hold code; CI holds execution.
2. **Different freshness:** Code updates daily; qTest export monthly; leadership asks weekly.
3. **Access fragmentation:** APIs blocked; teams use local exports.
4. **Metric blending risk:** Prior assessments combined implementation and execution — corrected in rebuild.
5. **No reconciliation owner:** Each team maintains local truth.

6. Implementation roadmap (aligned to 30/60/90)

Days 0–30

- Provision read-only credentials (Jira, qTest, GitLab)
- Extend `generate\_coverage\_intelligence\_assessment.py` for scheduled snapshots
- Publish v1 central register (CSV) with status labels
- Wire Mobile 2 GitLab nightly (QA-1405) — metric D/E
- JaCoCo delta pilot on `unite-mobile2` — metric A/E

Days 31–60

- qTest + Jira collectors live
- GitLab execution collector (pipeline last-run)
- `automation\_id` pilot on Mobile 2 tests (10+ linked)
- Reconciliation v2 with contradiction report
- Data-quality remediation sprint on qTest

Days 61–90

- GHA + Jenkins collectors
- Traceability soft gate on new QA automation MRs
- Required coverage check on pilot service
- Quarterly governance cadence approved
- Leadership reporting automation

7. Governance

| Element | Recommendation |

|-----|-----|

| Register owner | QA Automation Lead |

| Denominator approval | Product + QA Governance |

| Update cadence | Weekly automated; monthly leadership review |

| Freshness SLA | Execution metrics ≤ 7 days; inventory ≤ 30 days |

| Exception process | See code-coverage-gate-decision-brief.md |

| Contradiction SLA | Resolve or flag within 5 business days |

8. Explicit boundaries

| QA Automation (direct) | Other teams (required) |

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| Python collectors and register | Pipeline templates and required checks |

| Repo scans and normalization | GitLab/GitHub protected-branch rules |

| Leadership reports | JaCoCo/Sonar service configuration |

| Traceability standards | Jira scope hygiene, qTest cleanup |

| Pilot delta script | Bypass group, secrets, Jenkins API |

| Contradiction detection | Business criticality and risk tiers |

9. Success metrics (day 90)

| Metric | Target |

|-----|-----|

| Domains with documented denominator | ≥ 5 |

| Automated weekly register refresh | Yes |

| Repo → pipeline mapping | $\geq 80\%$ GS test assets |

| Jira ↔ qTest ↔ repo full chain (critical) | $\geq 30\%$ |

| Leadership metrics with formula + timestamp | 100% |

| Live API connectivity | GitLab + qTest + Jira Available |

Register authority: `government-savings-automation-assessment/03-analysis/verified-metrics-register.csv`